

IEOR 4404: Simulation

Lecture 14: Control Variate

Control Variate

- In estimating $E[g(X)]$, suppose in each replication we generate not only X , but also an “auxiliary” quantity Z
- This auxiliary quantity Z can be $Z = h(X)$ for some underlying random variate X .
- For convenience, call $Y = g(X)$. So each simulation run outputs (Y, Z) or $(g(X), h(X))$

Control Variate

- Suppose we know the information $E[Z] = \mu$
- Instead of outputting Y , we output

$$Y - \beta(Z - \mu)$$

where β is some chosen constant

- More generally, if we have n simulation runs, we output

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \beta(Z_i - \mu))$$

where (Y_i, Z_i) is the i -th simulation outcome

This is known as the control variate estimator, using control variate Z

Unbiasedness

Claim: The control variate estimator is unbiased

$$\begin{aligned} & E[Y - \beta(Z - \mu)] \\ &= E[Y] - \beta(E[Z] - \mu) \\ &= E[Y] \quad \text{since } E[Z] = \mu \end{aligned}$$

Unbiasedness

Claim: The control variate estimator is unbiased

Reasoning:

Consider one replication. The expectation is

$$E[Y - \beta(Z - \mu)] = E[Y] - \beta(E[Z] - \mu) = E[Y]$$

by the property of expectation

Like importance sampling, control variate is unbiased (however, we'll see some complications)

Control Variate

- If we can choose Z and β such that

$$\text{Var}(Y - \beta(Z - \mu)) < \text{Var}(Y)$$

then we achieve variance reduction

- Is this achievable, and if so what's the best way to attain this?

Variance Reduction

Observation 1: If we have full power to properly choose β , then control variate estimator has at most the variance of naïve Monte Carlo

Variance Reduction

Observation 1: If we have full power to properly choose β , then control variate estimator has at most the variance of naïve Monte Carlo

- We choose β such that the variance is reduced

$$\text{Var}(Y - \beta(Z - \mu)) = \text{Var}(Y) - 2\beta\text{Cov}(Y, Z) + \beta^2\text{Var}(Z)$$

Minimizing over β , we get $\beta = \frac{\text{Cov}(Y, Z)}{\text{Var}(Z)}$

- The variance is

$$\text{Var}(Y) - \frac{\text{Cov}(Y, Z)^2}{\text{Var}(Z)}$$

or equivalently

$$\text{Var}(Y)(1 - \rho^2)$$

where $\rho = \text{Corr}(Y, Z)$ is the correlation between Y and Z

Consider

$$\text{var}(Y - \beta(Z - \mu))$$

$$= \text{var}(Y) - 2\text{Cov}(Y, \beta(Z - \mu)) + \text{var}(\beta(Z - \mu))$$

$$= \text{var}(Y) - 2\beta \text{Cov}(Y, Z) + \beta^2 \text{var}(Z)$$

To minimize

$$\text{var}(Y) - 2\beta \text{Cov}(Y, Z) + \beta^2 \text{var}(Z)$$

we set derivative w.r.t. $\beta = 0$,

$$-2\text{Cov}(Y, Z) + 2\beta \text{var}(Z) = 0$$

$$\beta = \frac{\text{Cov}(Y, Z)}{\text{var}(Z)}$$

Choosing $\beta^* = \frac{\text{Cov}(Y, Z)}{\text{var}(Z)}$ gives the minimum var.

The minimum var is

$$\text{var}(Y) - 2\beta^* \text{Cov}(Y, Z) + \beta^{*2} \text{var}(Z)$$

$$= \text{var}(Y) - 2 \frac{\text{Cov}(Y, Z)}{\text{var}(Z)} \text{Cov}(Y, Z) + \left(\frac{\text{Cov}(Y, Z)}{\text{var}(Z)} \right)^2 \text{var}(Z)$$

$$= \text{var}(Y) - \frac{\text{Cov}(Y, Z)^2}{\text{var}(Z)}$$

$$= \text{var}(Y) (1 - \rho^2) \text{ where } \rho = \text{Corr}(Y, Z) = \frac{\text{Cov}(Y, Z)}{\sqrt{\text{var}(Y) \text{var}(Z)}}$$

If we choose the best β , i.e. $\beta^* = \frac{\text{Cov}(Y, Z)}{\text{var}(Z)}$,

then var of control variate estimator
(using one simulation replication) is

$$\text{var}(Y) (1 - \rho^2)$$

which is at most $\text{var}(Y)$.

Variance Reduction

The previous derivation also shows:

- **Observation 2:** As long as Y and Z have non-zero correlation, we can achieve variance reduction
- **Observation 3:** The more correlated (regardless of positive or negative) is Y and Z , the more variance reduction can be attained. In the most extreme case, if $\rho = \pm 1$, variance is 0.

To explain Observation 2,

if Y and Z have non-zero correlation, then $\rho \neq 0$, so

$$1 - \rho^2 < 1$$

Hence $\text{var}(Y)(1 - \rho^2) < \text{var}(Y)$

To explain Observation 3,

if Y and Z have corr 1 or -1,

then $1 - \rho^2 = 0$, so we have zero
variance for the control variate estimator.

In general, the higher is the correlation
(either sign), the more variance reduction

Intuition

Intuition 1: Control variate leverages the extra information on the mean of Z to reduce variance. However, this only works if Z is related to Y (in the sense of correlation).

Intuition 2: Control variate estimator is the intercept of a linear regression of Y against $Z - \mu$. Estimating this intercept has a lower variance than directly estimating the mean of the response variable.

the stronger the relation between Z and Y ,
the more variance reduction, as long as
we have the knowledge of $\mu = E[Z]$.

$$\beta^* = \frac{\text{Cov}(Y, Z)}{\text{Var}(Z)}$$

(linear regression:

$$Y = \alpha + \beta X$$

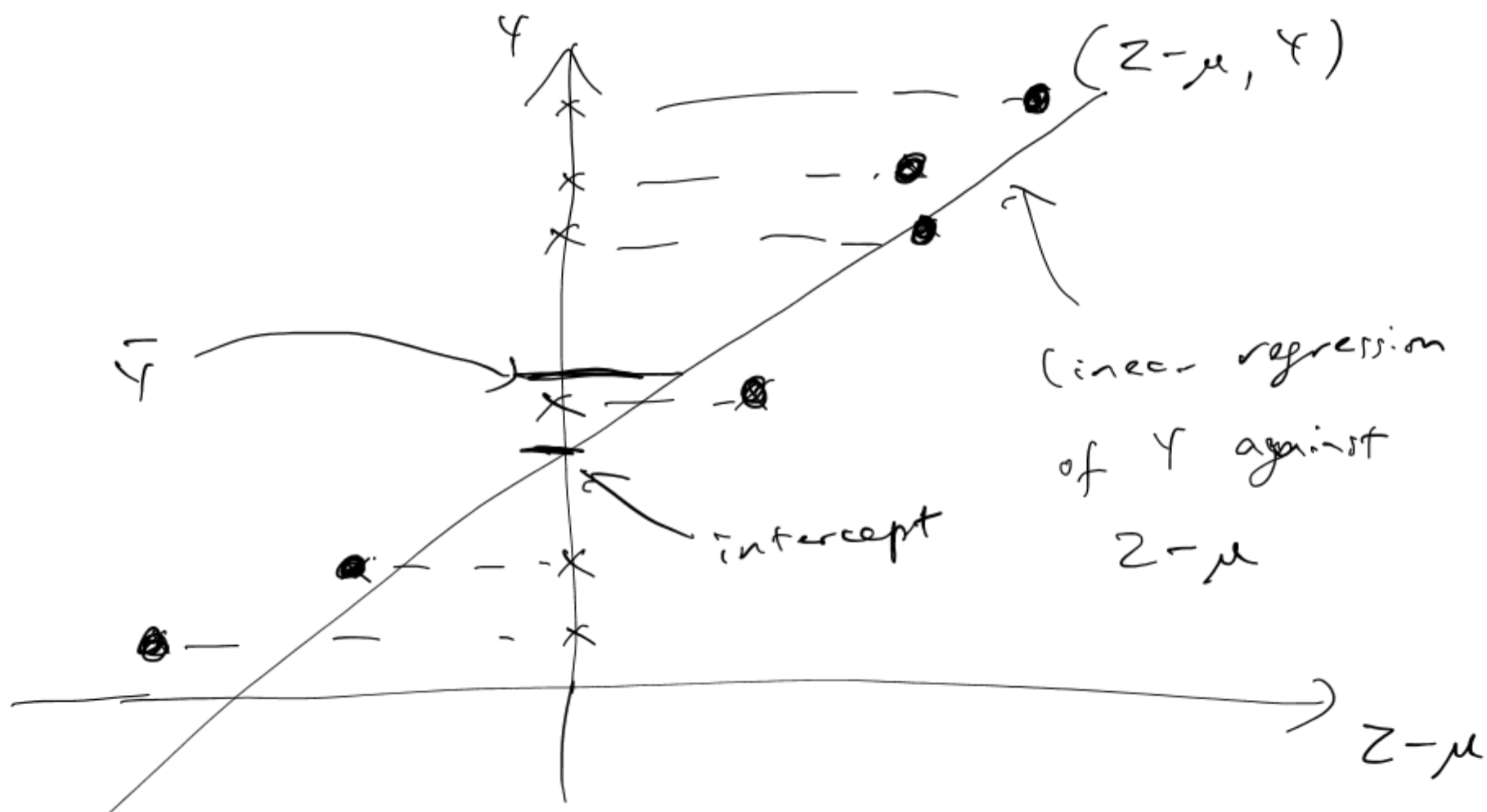
$$\hat{\beta} = \frac{\hat{\text{Cov}}(Y, X)}{\hat{\text{Var}}(X)}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

Control variate estimator:

$$\beta^* = \frac{\text{Cov}(Y, Z)}{\text{Var}(Z)}$$

$$\text{estimator} = \bar{Y} - \beta^* (\bar{Z} - \mu)$$



Multivariate Setting

- In practice, the optimal $\beta^* = \text{Cov}(Y, Z) / \text{Var}(Z)$ is unknown and estimated via its sample counterpart

$$\hat{\beta} = \widehat{\text{Cov}}(Y, Z) / \widehat{\text{Var}}(Z)$$

where $\widehat{\text{Cov}}(Y, Z)$ is the sample covariance between Y and Z , and $\widehat{\text{Var}}(Z)$ is the sample variance of Z , obtained from the simulation run itself

- Final output is

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}(Z_i - \mu))$$

This introduces a small bias (negligible relative to the standard deviation of the overall estimate)

$$E[Y - \hat{\beta}(Z - \mu)] \neq E[Y] - \underbrace{\beta(E[Z] - \mu)}_{=0} = E[Y]$$

because $\hat{\beta}$ depends on (Z_i, Y_i)

Multivariate Setting

All developments apply to multiple control variates: Suppose we have control variates $Z_1, \dots, Z_m$. Let's call them collectively $Z = (Z_1, \dots, Z_m)$. Suppose we know the information $E[Z] = \mu$ where μ is now a vector of means

- We output

$$Y - \beta'(Z - \mu)$$

where β is some chosen vector

- More generally, if we have n simulation runs, we output

$$\frac{1}{n} \sum_{i=1}^n \left(Y^{(i)} - \beta'(Z^{(i)} - \mu) \right)$$

where $(Y^{(i)}, Z^{(i)})$ is the i -th simulation outcome

$\nwarrow$
 $(\mu_1, \dots, \mu_m)$

$\nwarrow$
 $Y - \beta_1(Z_1 - \mu_1) - \beta_2(Z_2 - \mu_2) - \dots - \beta_m(Z_m - \mu_m)$

Multivariate Setting

- We choose β such that the variance is reduced

$$\text{Var}(Y - \beta'(Z - \mu)) = \text{Var}(Y) - 2\beta' \text{Cov}(Y, Z) + \beta' \text{Var}(Z) \beta$$

where $\text{Cov}(Y, Z)$ now denotes the covariance vector between the scalar variable Y and vector variable Z , and $\text{Var}(Z)$ is the covariance matrix of Z

Minimizing over β , we get $\beta = \text{Var}(Z)^{-1} \text{Cov}(Y, Z)$

- The variance is

$$\text{Var}(Y) - \text{Cov}(Y, Z)' \text{Var}(Z)^{-1} \text{Cov}(Y, Z)$$

- In practice, the optimal $\beta^* = \text{Var}(Z)^{-1} \text{Cov}(Y, Z)$ is unknown and estimated via its sample counterpart

$$\hat{\beta} = \widehat{\text{Var}}(Z)^{-1} \widehat{\text{Cov}}(Y, Z)$$

$$\text{var}(Y - \beta_1(Z_1 - \mu_1) - \dots - \beta_m(Z_m - \mu_m))$$

$$= \text{var}(Y - \beta'(Z - \mu))$$

$$= \text{var}(Y) - 2 \beta' \underbrace{\text{Cov}(Y, Z - \mu)}_{\substack{\uparrow \\ (Z_1, \dots, Z_m)}} + \beta' \underbrace{\text{var}(Z)}_{\substack{\downarrow, \text{Covariance matrix}}} \beta$$

$(\beta_1, \dots, \beta_m)$ $(\beta_1, \dots, \beta_m)$
 $\downarrow$ $\downarrow$

$$\begin{matrix} (Z_1, \dots, Z_m) & (\beta_1, \dots, \beta_m) & (\text{Cov}(Y, Z_1 - \mu_1), \dots, \text{Cov}(Y, Z_m - \mu_m)) \\ \downarrow & & \\ \text{var}(Z) = & \left(\begin{array}{ccc} \text{var}(Z_1) & \text{Cov}(Z_1, Z_2) & \dots \\ \text{Cov}(Z_1, Z_2) & \text{var}(Z_2) & \dots \\ \vdots & \vdots & \ddots \\ \dots & \dots & \text{var}(Z_m) \end{array} \right) \end{matrix}$$

Take derivatives and set to 0,

$$\beta^* = \underbrace{\text{var}(Z)}^{-1} \text{Cov}(Y, Z)$$

$$\begin{pmatrix} \text{var}(Z_1) & \text{Cov}(Z_1, Z_2) & \dots \\ \text{Cov}(Z_2, Z_1) & \text{var}(Z_2) & \\ \vdots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \text{var}(Z_m) \end{pmatrix} \begin{pmatrix} \text{Cov}(Y, Z_1) \\ \vdots \\ \text{Cov}(Y, Z_m) \end{pmatrix}$$

In the case of 2 control variates,

$$\beta^* = \begin{bmatrix} \text{var}(z_1) & \text{cov}(z_1, z_2) \\ \text{cov}(z_2, z_1) & \text{var}(z_2) \end{bmatrix}^{-1} \begin{bmatrix} \text{cov}(Y, z_1) \\ \text{cov}(Y, z_2) \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \widehat{\text{var}}(z_1) & \widehat{\text{cov}}(z_1, z_2) \\ \widehat{\text{cov}}(z_2, z_1) & \widehat{\text{var}}(z_2) \end{bmatrix}^{-1} \begin{bmatrix} \widehat{\text{cov}}(Y, z_1) \\ \widehat{\text{cov}}(Y, z_2) \end{bmatrix}$$

control variate estimator

$$= \frac{1}{n} \sum_{i=1}^n \left[Y^{(i)} - \beta' \begin{bmatrix} Z_1^{(i)} - \mu_1^{(i)} \\ Z_2^{(i)} - \mu_2^{(i)} \end{bmatrix} \right]$$

Example

G/G/1 queue:

The goal is to estimate the expected sum of waiting times, say $E[\sum_{i=1}^{N(t)} W_i]$ where $N(t)$ is the number of arrivals before time t , and W_i is the waiting time of the i -th customer

Suppose we know the mean of the service times S_i . We can use control variate $\sum_{i=1}^{N(t)} S_i$.

naive MC:

• generate

$$\sum_{i=1}^{N(t)^{(j)}} W_i^{(j)} \quad \text{for } j = 1, \dots, n$$

(use Lindley's recursion)

• Output

$$\frac{1}{n} \sum_{j=1}^n \sum_{i=1}^{N(t)^{(j)}} W_i^{(j)}$$

Control variate estimator:

$$\cdot \text{Generate } \left(\sum_{i=1}^{N(t)^{(j)}} \omega_i^{(j)}, \sum_{i=1}^{N(t)^{(j)}} S_i^{(j)} \right)$$

for $j=1, \dots, n$

$$\begin{aligned} \text{mean of control variate} &= E \left[\sum_{i=1}^{N(t)} S_i \right] \\ &= E[N(t)] E[S] \end{aligned}$$

• Output

$$\frac{1}{n} \sum_{j=1}^n \left(\sum_{i=1}^{n(t_j)} w_i^{(j)} - \hat{\beta} \left(\sum_{i=1}^{n(t_j)} S_i^{(j)} - E[n(t_j)] E[S] \right) \right)$$

where

$$\hat{\beta} = \frac{\widehat{\text{Cov}} \left(\sum_{i=1}^{n(t)} w_i, \sum_{i=1}^{n(t)} S_i \right)}{\widehat{\text{Var}} \left(\sum_{i=1}^{n(t)} S_i \right)}$$